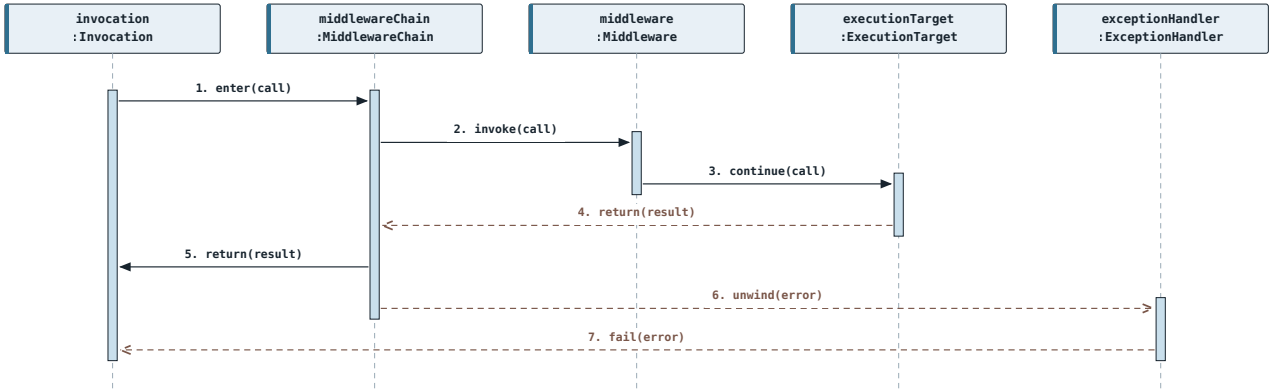
P04 Middleware-pipeline loop

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Layer behavior around model/tool calls.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A harness surrounds model or tool invocation with extensible behavior.
TRIGGER	A call enters or leaves the runtime boundary.
STRUCTURE	Ordered middleware entries with a continuation or short-circuit result.
EFFECT	Logging, policy, retry, or context behavior runs in declared order.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	The boundary exposes a composable continuation contract.
ENSURES	Middleware executes in deterministic order and the final result reaches the loop.
INVARIANT	A middleware does not silently bypass required downstream layers.
VIOLATION	Short circuits and exceptions unwind deterministically with one terminal result.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).
Analytical collaboration only; no repository execution or message order is asserted.